AI-BASED FINANCIAL REPORT ANALYSIS SYSTEM

Mini Project Report

Project focus: Upload-driven financial PDF analysis with a trained machine-learning revenue forecasting model.

Prepared as a runnable academic mini-project package

Technology stack: Python, Streamlit, pdfplumber, Pandas, scikit-learn, Plotly, ReportLab

Abstract

This project implements an AI-assisted system for analyzing financial reports supplied as PDF files. The Streamlit interface extracts selectable text from a user-uploaded report, parses common financial labels and values, and lets the user review or correct the extracted data before analysis. The application calculates financial ratios, compares periods, flags potential anomalies, generates a baseline trend forecast, and exports a PDF analysis report. A Random Forest model is trained on company-year financial facts prepared from the SEC Financial Statement Data Sets supplied for this project. The trained model forecasts next-period revenue using Revenue, Net Profit, and Total Assets. Crucially, training data is kept separate from report analysis: the app does not populate missing PDF values from the training dataset.

1. Introduction

Financial reports contain information about revenue, profit, assets, liabilities, liquidity, and performance trends. Manually locating these values and comparing periods can be time-consuming. This project automates the first stage of analysis while keeping a human verification step, because PDF layouts and financial terminology differ between organizations.

2. Problem Statement

Develop a small, locally runnable application that accepts financial PDFs, extracts supported financial values, performs transparent calculations and visualizations, optionally uses a trained model for revenue forecasting, and produces a downloadable analysis report without displaying predefined demo financial figures.

3. Objectives

- Extract selectable text from an uploaded PDF.
- Parse common financial metrics and period labels.
- Let users review and edit extracted values before analysis.
- Calculate financial ratios and period-to-period changes.
- Identify potential outliers and provide a revenue forecast.
- Train and integrate a machine-learning model using real reported financial facts.
- Export a PDF report and provide reproducible setup instructions.

4. System Architecture

1. User Uploads financial PDF	**2. PDF Parser** Text extraction + metric parsing	**3. Review Table** User verifies extracted values
4. Analysis Engine Ratios, comparisons, anomalies	**5. Forecasting** Baseline trend + trained model	**6. Output** Dashboard + PDF report

The training pipeline is separate from the PDF upload pipeline. SEC data trains the model offline; uploaded PDF values are the only values used for the current report analysis.

5. Tools and Technologies

Python is used for application logic and model training. Streamlit provides the user interface. pdfplumber extracts selectable PDF text. Pandas and NumPy support data preparation and analysis. scikit-learn supplies the RandomForestRegressor and IsolationForest. Plotly displays interactive metric trends. ReportLab generates downloadable PDF reports. joblib serializes the trained model.

6. Dataset Preparation

The user-provided SEC quarterly archives covered 2022 Q1 through 2025 Q4. The preprocessing script filtered annual 10-K/10-K/A submissions, selected common US-GAAP tags, and assembled a company-year table. SEC archives can contain comparative facts for earlier fiscal years, so the resulting table includes periods from 2007 to 2025. The generated training CSV contains 41,650 company-year rows from 7,541 companies before the model-specific completeness and positive-revenue filters.

7. Model Training

The target is next-period revenue for the same company. Input features are current-period Revenue, Net Profit, and Total Assets. A Random Forest regression model is trained after signed-log transformation of features and log1p transformation of the positive revenue target. A company-held-out split is used where enough companies are available, so companies in the test split are not included in the training split. The model is serialized to models/revenue_forecast_model.joblib.

8. Training and Validation Results

Metric	Result
Usable next-period examples	17,060
Companies represented in training examples	5,167
Held-out test rows	3,401
Model	RandomForestRegressor
Test R^2	0.9743
Test MAE (USD)	$844,566,115
Test RMSE (USD)	$4,554,027,426
Median absolute percentage error	16.13%
Baseline current-revenue MAE (USD)	$905,464,084

Interpretation: the test-set R^2 is high, but the revenue distribution is highly skewed across companies. The mean percentage error is strongly affected by small-denominator cases, so the median percentage error and USD error metrics should also be considered. The baseline current-revenue MAE is shown as a simple reference; model performance should not be described as guaranteed accuracy.

9. Financial Analysis Features

- Liquidity: Current Ratio = Current Assets / Current Liabilities.
- Leverage: Debt-to-Equity = Total Debt / Shareholders' Equity.
- Profitability: Net Profit Margin = Net Profit / Revenue × 100.
- Return on Assets: Net Profit / Total Assets × 100.
- Period comparison: absolute and percentage change between available periods.
- Anomaly detection: flags unusual observations within each metric when sufficient values exist.
- Forecasting: trained revenue prediction when all required metrics are extracted; otherwise a transparent baseline forecast may be used.

10. User Workflow

1. Start the app with Streamlit.
2. Upload a text-based financial PDF.
3. Inspect extracted PDF text if needed.
4. Review and edit the candidate values table.

5. Click Analyze this PDF.
6. Review ratios, comparisons, anomaly flags, forecasts, and insights.
7. Download the PDF analysis report.

11. Limitations

The parser is label-based and may not correctly interpret every PDF table layout, footnote, currency scale, or accounting convention. Image-only scanned PDFs require OCR, which is not enabled. Extracted numbers must be checked by the user. The trained model reflects SEC-reporting companies and may not generalize to private companies or Indian financial statements. Financial forecasting is approximate and must not be treated as investment, accounting, or tax advice. The project is a mini-project prototype, not a production audit tool.

12. Future Enhancements

Add OCR for scanned PDFs, more robust table extraction, currency and unit normalization, a larger taxonomy of financial statement tags, sector-aware models, temporal backtesting, explainability tools, and support for Indian company filings. A production version should add validation logs, test coverage, access controls, and stronger data governance.

13. Conclusion

The project demonstrates an end-to-end workflow for PDF-based financial analysis and a trained revenue forecasting model. It keeps uploaded-report analysis separate from the training corpus, includes model validation metrics, and provides a runnable Streamlit interface with downloadable output. Human review remains essential because financial PDFs vary significantly in structure.

Dataset source context: SEC Financial Statement Data Sets (DERA), provided by the project user. The processed CSV and training metrics are included in this package.